Kausthub Kannan

9353633373 kausthubkannan961@gmail.com Linkedin GitHub Kaggle

Building production-grade AI agents, intelligent document processing systems, and scalable ML solutions— Japanese (N5)

Experience

Associate IT Engineer **August 2025 – Present**
Maximus *Bangalore*

- Built a modular document processing platform and SDK integrating AWS Textract, managed inference services, multiple OCR engines, and LLMs, enabling complexity-aware engine orchestration that improved extraction efficiency by 72%, reduced manual processing errors, and lowered document processing costs by up to 68%.
- Automated deployment of quantized open-source LLMs on monitored Amazon EC2 instances by creating LLM-optimized AMIs, enabling scalable, reproducible inference infrastructure for document classification workloads while reducing deployment time and resource utilization.
- Engineered an XAI validation system using a custom NLI-based explainability framework for LLM-generated data and summaries, improving model reliability and output transparency while providing technical guidance to interns during development.

Software Engineering Intern **February 2025 – July 2025**
Maximus *Bangalore*

- Built automated document processing pipelines using AWS Textract; benchmarked system performance across extraction accuracy, latency, and cost metrics.
- Modularised an evaluation framework into plug-and-play serverless Lambda blocks, enabling reuse across different pipeline architectures without code changes.

Research and Development Intern **March 2024 – November 2024**
Siemens Advanta *Bangalore*

- Reduced memory footprint on edge panel devices via model quantization and embedding compression; delivered a deployment SDK for constrained hardware environments.
- Fine-tuned open-source LLMs for HMI environments, improving QA accuracy in low-resource settings with limited compute budgets.

Projects

Poneglphy - Open Source Plugin to enable Agents as Research Brains **Github**

- Built an open-source Obsidian Community Plugin used by **500+ active users** that converts research hypotheses into citation-backed knowledge graphs using autonomous deep-research agents.
- Engineered a recursive multi-agent framework with vector search, semantic retrieval, and HITL verification, generating topic analyses with multiple cited sources while reducing manual research time by approximately **85%**.

Tachyonot — Open Source LLM compiler specific to system architecture **Github**

- Engineered an edge-AI compiler for running quantized LLMs and Retrieval-Augmented Generation (RAG) workloads on AVX-capable CPUs, delivering practical offline inference without dedicated accelerators via building system specific binaries.
- Reduced deployment requirements to **less than 1 GB runtime memory** and **1–2 GB storage** through optimized model loading, retrieval pipelines, and resource-aware execution, enabling operation on older consumer systems.

Publications & Certifications

DROPEX: Disaster Rescue Operations and Probing using EXpert drones **CSITSS IEEE 2025**
DOI: 10.1109/csitss64042.2024.10816931 *Bangalore, Karnataka*

AWS Certified Machine Learning Engineer – Associate **2026**
Certificate Credentials *Validity: 3 years*

Technical Skills

Languages: Python, TypeScript, SQL, Bash
Cloud & Infra: AWS (Textract, Bedrock, Step Functions, SageMaker, Lambda, CFT, Glue), Azure, Docker
ML & Agentic AI: PyTorch, OpenCV, Langgraph, LiteLLM, AWS Agentcore, Deepagents, MLOps, MLFlow, ML Pipeline Deployment & Monitoring
Backend: FastAPI, Flask, REST APIs, Postgres

Education

Dayananda Sagar College of Engineering **December 2021 – June 2025**
BE in Computer Science and Design – 9.6 CGPA *Bangalore, Karnataka*